DeepSynBa Project Report

Small data reproduction and model experiments

Prepared for Professor Oznur Tastan | 9 October 2026

Project github.com/Kami80/DeepSynBa | Work completed through 8 October 2026

I reproduced your team's released DeepSynBa implementation on my laptop and investigated its behavior with limited training data. My contribution is a reproducible local workflow, model diagnostics, and controlled experiments on training setup and parameter outputs. The enhanced variant reduced fresh-pair holdout RMSE by **32.4%** relative to my local reference under the same small-data setup. [3]

Work completed

I prepared a deterministic subset of **5,000 complete 4 x 4 response matrices** from the released NCI-ALMANAC archive: 3,000 for training, 1,000 for validation, and 1,000 for the original test. This uses 1.82% of the archive's 274,516 matrices. I preserved the original partition of unordered drug pairs, checked for overlap and duplicates, and retained all dose entries and response units. The subset covers all 101 drugs and 60 cell lines in each partition. [3]

I added a Windows-compatible plain PyTorch runner with local logging, fixed seeds, early stopping, validation-based checkpoint selection, and saved predictions. The reference preserves the released model, precomputed MolFormer and gene-expression features, response equation, MSE loss, and AdamW optimizer. I also corrected batch-dimension handling for single-example inference and checked checkpoint reloads and data integrity.

Three-seed reference runs revealed zero-valued Hill-slope outputs and sensitivity to initialization. I then compared **10 configurations across three seeds**, testing positive-output activation with near-one initialization, smaller network widths, drug-order augmentation with corresponding dose and target transposition, and bias ablations. Follow-up studies added fresh-pair confirmation, 27 nested-data fits, 18 optimizer-budget fits, and 12 activation–initialization fits. [3]

Comparison with the original project

Aspect	Original DeepSynBa [1, 2]	My local study [3]
Data and scope	Large NCI-ALMANAC and O'Neil evaluations.	5,000 NCI-ALMANAC matrices, with 3,000 used for training.
Generalization	New drug combinations, cell lines, and drugs; synergy analyses.	New drug pairs among known drugs and cell lines; an additional fresh-pair holdout.
Contribution	Original response-surface model and broad benchmark comparisons.	Local reproduction, head diagnostics, compact variants, and controlled optimization studies.

For context, the official repository reports new-drug-combination RMSE **7.72**, Pearson **0.95**, and Spearman **0.82**. [2] My scores use much less training data and a different evaluation scope; they quantify improvements over my local reference and do not reproduce the full-data benchmark.

Selected experimental results

The three models below use the same 3,000 training matrices and seeds 42, 43, and 44. Configurations were selected by mean validation error. [3]

Model	Parameters (millions)	Original test RMSE	Fresh pair holdout RMSE
Released-code reference	40.94	21.327 ± 2.095	21.357 ± 2.247
Positive outputs plus drug-order augmentation	40.94	14.605 ± 0.390	14.443 ± 0.367
Compact model plus drug-order augmentation	3.50	15.161 ± 0.082	15.019 ± 0.011

Lower RMSE is better. Values are mean ± sample SD across three seeds. Each holdout has 1,000 complete matrices; RMSE pools all 16,000 dose entries on the original response scale, with untreated reference 100.

The fresh holdout contains **323 previously unused drug pairs**, excluding every pair in the earlier subsets. Frozen enhanced checkpoints achieved RMSE **14.443 ± 0.367** versus **21.357 ± 2.247** for the reference: a 32.4% reduction (paired drug-pair bootstrap 95% interval 29.0–35.8%, conditional on the frozen models). The compact variant uses **91.5% fewer parameters**, with fresh-holdout RMSE 15.019. [3]

Findings from the controlled studies

Nested training sizes of 500, 1,000, 2,000, and 3,000 matrices showed that training rules affect the apparent cost of reducing data. With a common maximum of 2,400 optimizer updates and update-based scheduling and stopping, the enhanced 2,000-matrix model achieved validation RMSE **14.343 ± 0.178**. This is 0.294 above its 3,000-matrix mean while using one third fewer training matrices. This comparison uses validation data only. [3]

The final 2 x 2 study separated activation from near-one output calibration at 2,000 training matrices. Original ReLU gave validation RMSE **20.378 ± 2.132**; calibrated ReLU gave **14.408 ± 0.091**. Calibrated Softplus gave **14.343 ± 0.178**, only 0.065 lower than calibrated ReLU. These results show that initialization is an important part of the improvement; the earlier gain cannot be attributed to the activation change alone. [3]

Scope and next step

The work establishes a functioning small-data reproduction and evidence for improved local variants. The original test was reused during development, the fresh holdout came from the same experimental source, and the later optimization studies reused validation data. Three-seed SDs describe training variability. External-source validation, unseen-drug/cell evaluation, competing-model retraining, and pharmacological parameter validation remain open. The next evaluation should freeze the final comparison and use a reserved holdout or independent dataset.

References and project records

[1] Kuru HI, Zhang H, Rattray M, Ek CH, Cicek AE, Tastan O, Milo M. DeepSynBa: actionable drug combination prediction with complete dose-response profiles. Bioinformatics 42(7), btag416 (2026). doi.org/10.1093/bioinformatics/btag416

[2] Original code and repository benchmark: github.com/hikuru/DeepSynBa (released commit 43c1696).

[3] My source, protocols, per-seed results, and analysis summaries: github.com/Kami80/DeepSynBa/tree/main/results.